The Keynes Theory, ch.5

Not really, what did you expect?

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Keynes's theory of employment, if you don't expect something big happen and exciting you usually do, I don't know, it can turn on a pin.

– The Books, "Intelli Life Centre"

To this it, the same we or clerm people, the same ten or clerm partier, the same ten or clerm rung, playing out on an radius loop in an radius room that's a decade long,... But will I end up naked on a highway, somewhere in new jersey somewhere in japan? how did I get here? how did these pills get from Nicaragua into my hands?

– Long Long Long, "If There's A Bumble You Ought A Rock Me, Right?"

Gambling discourses has thankfully mostly passed me, aside from whatever ends up in my Facebook feed. Nevertheless, the Keynes Theory of Employment, as one of the microeconomic concepts I am important, because the will prove useful in my future series on Ch.5 Intro today. The plan is to use the language of order book depth to talk about the way Keynes uses "expectation" in the General Theory.

Chapter 5 is short and sweet Keynes lays out what he means by "expectation" and the basics of how these expectations shake out into economic activity. It's rough though, because expectations... how do you measure expectation? People like to try with sentiment surveys and natural language processing and word counters, but in Keynes time, it was both more obscure and more straightforward. Less information flying around meant people had to rely on their hands and feet to see what was actually happening, but a much better handle on what information other participants had. Lost an – in degree of 10 and like to read ahead – I don't know how why the latest kind of information matters more than the former. But for now, Keynes is just setting out how to think of "expectation" as a concern, if not quantifiable concern.

For Keynes, expectation is basically the medium through which ideas pass in order to be incorporated into the economy. As a medium, it exists in two forms as ideas, and it is fully possible to succeed by understanding the medium and not understanding the ideas underneath it. Keynes's Ch.5 is a little bit of a philosophical discussion of the nature of economic activity and entrepreneurs as coming from "the discussion in every" and "the decision in every" coming from expectation, it would really helpfuns to us have a good handle on how expectations are meant to work in the underlying world.

How Probable Are Your Probabilities?

There are two basic questions we are going to cover today and they are: What does Keynes mean by the concept of expectation? And where do that expectation get that makes them matter for the economy as a whole?

Before the first question, it's helpful to start out at Keynes' Treatise on Probability. One of the central animating questions of all of Keynes' work is "how is it possible to act under uncertainty and then, how is it possible to act successfully?" It's important that we study Keynes on this, because otherwise action under uncertainty and decision theory gets split in ten dozen ways with BANS Guy brain and so on and so forth.

In the Treatise, actors form beliefs, but also beliefs about the viability of those beliefs. Maybe an actor wants to put it to that belief to the likelihood of an event happening, and then firmly one holds that belief, are in principle separable quantities. This is a little bit different than the basic between frequentists and Bayesians where probability is meant to answer the question "what happens on average" or "what's going to happen to me?", but not too far afield.

This structure of beliefs and meta-beliefs that Keynes begins with eventually gives rise to a whole weird metaphysical/epistemological literature that we are going to go into here. We're having a nice time, and really wanted to think about "non-additive probabilities", or "subnegs integrals" or any of that. In the middle even that you do, drop a line in the comments and I can point you towards the work of Reid O Donnell and Anna Conybeare and the rest of them.

This is all pretty abstract, so, for a first example, let's take the weather. You might say, before checking the weather, that you think it's probably going to rain tomorrow. You're not very attached to that estimate, because you haven't checked the weather, but you have some kind of belief about what's likely. Then, tomorrow comes and you look like what they said, maybe, but you also find a little bit like it's called for snow, and "like a little bit" is pretty low. So you say, maybe there's a 40% chance it rains tomorrow, and you assign a much greater maybe 50%. At this point, you think it's 50% whether there's a 40% chance of rain tomorrow, or some unspecified other probability. But then you watch the weather report and it says 60% for sure that it's going to rain. Different

Now, you know meteorologists get it wrong a lot of the time, so you don't think a 100% chance of rain tomorrow is a sure thing. Maybe you then assign a 70% chance that the weatherman is right, and a 75% chance he's wrong. But, just because you think the weatherman might be wrong, doesn't mean you believe that he's wrong 75% of the time that he is right. So you add your 50% belief to a 40% chance of rain to your 25% belief in a 100% chance of rain.

What's critical about this whole operation is that belief's accrete to people through interaction with information that the world provides. This leads little off by something library like "will it rain tomorrow," but is actually good epistemic practice. The future is uncertain, so there is no reason to believe any agent knows what absolutely all of the possible outcomes look like. So rather than making iterative guesses of what outcomes you think for sure will happen, you adjust your different information sources weighed by how much you believe them.

In the above example at the end you have two beliefs. One is that there is a 40% chance of rain tomorrow, and another is a 100% chance of rain tomorrow. You are weighing the first belief at 75% and the second belief at 25%. Where does that 75 and 25 come from? Well, weighing is crazy, but "another 40% chance of rain tomorrow" is crazy, too? This makes it impossible to get an exact estimate of your expectation by setting the weightings, because the weightings don't seem to it. I think you're clearly evaluated every possible possibility, some amount of your weighting has no context.

Here, this looks a little silly, almost a bizarre event like "will it rain tomorrow" the weightings don't matter because the probabilities are independent, and the outcomes are completely and open-ended. There's no reason to assume that at any given point you can give an exact probability distribution for the future, because no serious observer ever really has as though they have fully evaluated every possible event variable about a future event.

This is problematic, and is what we've talked about before, about the difference between debunking and documenting. Keynes' approach to probability here is really an account of what documenting the New Subject looks like. Rather than assuming one knows the whole probability space, and wanting to be defeated to change opinions, the Treatise's approach, which continually adds more information and adjusts its weighting appropriately. Rather than rational, top down evocanes, beliefs and beliefs about beliefs are accumulated through experience. But this probability structure is also – and this will play an important role later on – a fundamentally lazy system. Rather than constantly updating every belief, and every belief about a belief, one will go right on thinking that the future will be basically like the past.

Just as an aside, this is subtly different from Tabb's arguments about epistemology and probability. Tabb's are mostly those of Frank Knight, but expressed a little differently. Knight's idea is that, for a lot of things, there just isn't enough or the particular kind of observation of a given time series to be sufficient to determine the range of values is taken. The world has a way that it is – specified probability distribution – but we can never finally finish observing it. We just have to make guess from arguments about Keynesian fundamental uncertainty another day.

What Can You Expect From Your Expectations?

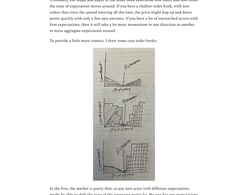
This framework for thinking about probabilities pays out pretty smoothly in thinking about expectations. We just talked in our last piece about how it's basically impossible to pick down a whole pile of subjective estimates and assign them weights, so why is Keynes doing exactly that here with Expectation, literally an aggregation of subjective estimates?

To talk about the "state of expectation," it's easiest I think to compare it to the order book in a given market. Here, granted, the way this all unfolds in Keynes is more complicated and keep bore nowadays, but the principles are basically the same.

The basic idea behind an order book is to pile up everyone's moving limit orders, and execute when two orders match. You have one group of people saying "I will buy some quantity of X at some specific price Y, if anybody wants to sell me Y, and other people offering to sell some quantity of X at price Z, if anybody wants to buy it from them at price Z." When a buyer and seller agree on a price, a trade gets made. Over time, Wherever a trade happens, the price is happy that it's happened, and the rest of the world can see a new future gets set to be at that price. If there's no point of overlap, traders don't happen and everyone's orders sit there until some change occurs. It'll then orders show up past the limits of existing orders on one side or the other, then The Price moves. This is the same fundamentally lazy structure as in our probabilities explanation above: unless there's a good reason for price to change too, new orders will stay the same.

In some sense, The Price stands in symbolically for the order book process that creates The Price. The "state of expectations" resides in the same way. Individual actors set up expectations anywhere in the order book, and the "state of expectations" that one observes is like a low-resolution price.

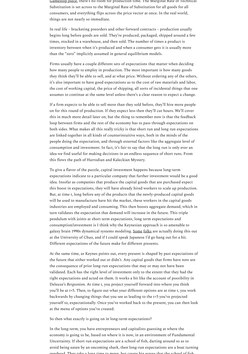
To provide some visual grounding, order books look basically like this:



Now, it's easier to think of the social process that generates the market's "state of expectation" in Keynes' system as being basically shaped like an order book. Rather than price and size, you have expectation and uncertainty. If you wanted to take a "skin in the game" approach, you'd say commitment is based on how much cash one had invested in some expectation. You could also be a Keynesian thing, and consider the commitment to be an investment of social capital, but we all know how shitty that is, especially in economic and finance.

Ultimately, the shape and depth of the order book determine how much and how often the state of expectation moves around. If you have a deeper order book, with new orders that cross the spread entering all the time, the price might hop up and down pretty quickly with only a few new orders, but if you have a lot of entrenched actors with firm expectations, there will take a lot more momentum in one direction or another to move aggregate expectation around.

To provide a little more context, I drew some case order books:



In the first, the market is pretty thin, so any new actor with different expectations might be able to shift the state of the aggregate pretty far. We see that are expectations that they might potentially hearth, and so they are very open to revision and permanent. In the second, there's little room to run to the right, but not so far left. This is a market where people think that the price is basically determined, but high-bread risk-prone is certain. In the last one, things are basically set in the left, having some really extreme new information. But once they start moving to the left, there's little resistance, the world could be anything.

This whole structure also makes possible – and we'll go into this in greater depth in Chapter 12 – arbitrage opportunities, as well as self-fulfilling prophecies. Basically, the assumption is that everyone at home calculates their own expectation and commitments and then arbitrage them to market. However, some agents can look back at average opinion reflectively, and begin to think about payoffs based on the expectations of other rather than how they expect the world to go. If you look at the order book of expectation, and have a house guess as to how it's going to move, you can position yourself based purely on that, rather than a fundamental expectation about what the future state of the world are going to be. This is how contrarianism works, and it's how Keynes's system allows for it, too. Keynes's system allows for speculation based on expected expectations, it's sometimes possible to "make dech happen" purely by getting enough expectational juice behind it. But we'll get to all of that later. If you want to chase that, I recommend reading Capital and The Martian Keynes.

What Did You Expect Your Expectations to Do?

Now that we've explained what Keynes means by expectations, the next step is to explain what that do. Keynes distinguishes between short-run and long-term expectations, and analyzes each separately. Short-term expectations are really about employment, while long-term expectations are about capital formation.

This makes sense on the short end: it takes some time to produce stuff, but not that much time. It's helpful to have an expectation of the economy that I'm able to sell and at what price. With an existing capital stock, you can usually vary how many people work on a given widgets, and move businesses to try to buy stuff to sell at a profit